

Assignment 3

CS543 – Web Services Assignment 3

Integrate an External SOAP Partner

Team ID: Group_14

Team Members:

- 1)** Shivam Kumar (20251651083)
- 2)** Aman Tripathi (20251651018)
- 3)** Raghav Soni (20251651076)
- 4)** Nikhil Singh Tomar (20251651065)
- 5)** Aniket Kumar (20251651019)

1) Context

CampusEats integrates with UniPay Bank Payment Gateway to process payments for student food orders. The selected operation is charge, which is responsible for authorizing and collecting payment before an order is confirmed.

Although most CampusEats services communicate through REST APIs, the payment integration uses SOAP because the banking partner already exposes a SOAP-based contract. In financial systems, reliability is more important than simplicity. SOAP provides a formal WSDL contract, message-level security support, and transaction-oriented communication, making it suitable for handling sensitive payment requests. Therefore, CampusEats keeps its internal services RESTful while using SOAP specifically at the banking integration boundary.

CampusEats did not choose SOAP for convenience; it uses SOAP because interoperability with the banking ecosystem requires adherence to the partner's contract and standards.

2. Partner WSDL

The complete WSDL contract for the UniPay Payment Gateway has been provided separately in the file partner.wsdl. The contract defines the data types, messages, operations, SOAP binding, and service endpoint required for integration.

3. SOAP Messages

The SOAP request, SOAP response, and SOAP fault messages have been provided as separate XML files:

- soap-request.xml
- soap-response.xml
- soap-fault.xml

These files demonstrate the request, successful response, and fault handling scenarios for the charge operation.

4. HTTP Binding

The following HTTP request carries the SOAP envelope for the charge operation.

POST /payments HTTP/1.1

Host: api.unipay-campus.example

Content-Type: text/xml; charset=utf-8

SOAPAction: "urn:unipay:payments/charge"

<?xml version="1.0" encoding="UTF-8"?>

<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"

xmlns:uni="urn:unipay:payments">

<soap:Header>

<uni:auth>

<merchantId>MERCHANT_CS543_001</merchantId>

<apiKey>a1b2c3d4e5f6g7h8i9j0</apiKey>

</uni:auth>

</soap:Header>

<soap:Body>

<uni:chargeRequest>

<merchantId>MERCHANT_CS543_001</merchantId>

<cardToken>tok_student_9f2c4a8b</cardToken>

<amount>299.50</amount>

<currency>INR</currency>

<orderReference>ORD_CAMPUS_2026_001</orderReference>

</uni:chargeRequest>

</soap:Body>

</soap:Envelope>

5. Discovery and Registry Entry

CampusEats discovers the UniPay Payment Gateway through a service catalogue maintained by the university IT department. When CampusEats requires a payment service, it performs a registry lookup and obtains the service endpoint and WSDL location. The WSDL is then downloaded and used as the contract for integration. Alternatively, the partner also provides a direct WSDL URL for consumers that already know the service location.

Field	Value
Business	UniPay Bank Ltd. (Campus Banking Division)
Service	UniPay Payment Gateway
Endpoint URL	https://api.unipay-campus.example/payments
WSDL Location	https://api.unipay-campus.example/payments?wsdl
Protocol	SOAP 1.1 / HTTP
tModel	urn:unipay:payments
Operations	charge, refund
Status	Active

6. Fault Mapping

CampusEats translates UniPay SOAP faults into its own internal error codes before returning responses to users. This prevents bank-specific terminology from leaking into the user interface and ensures consistent error handling across the platform.

UniPay Fault	CampusEats Error
CARD_DECLINED	PAYMENT_FAILED
INSUFFICIENT_FUNDS	INSUFFICIENT_BALANCE
EXPIRED_CARD	PAYMENT_METHOD_EXPIRED
INVALID_TOKEN	INVALID_PAYMENT_METHOD
GATEWAY_TIMEOUT	PAYMENT_TIMEOUT
GENERAL_ERROR	INTERNAL_ERROR

Example Error Translation:

```
{  
  "errorCode": "PAYMENT_FAILED",  
  "message": "Payment declined. Please try a different payment method or  
contact your bank.",  
  "timestamp": "2026-08-16T14:35:22+05:30"  
}
```